



LHE

Environmental
Hydraulics
Laboratory

Including hyporheic flow in the vertical velocity profile

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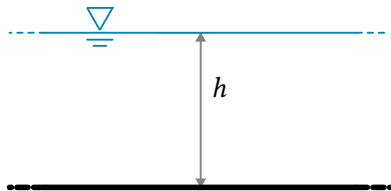
25.03.2026

Flow resistance

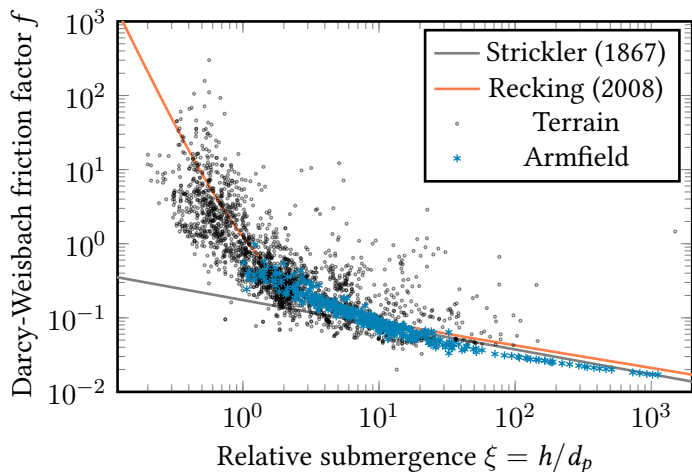
According to the (Gauckler-)Manning-Strickler formula, the bed's shear resistance is :

$$\tau_p = \frac{\rho g}{K^2} \frac{\bar{u}^2}{h^{1/3}}$$

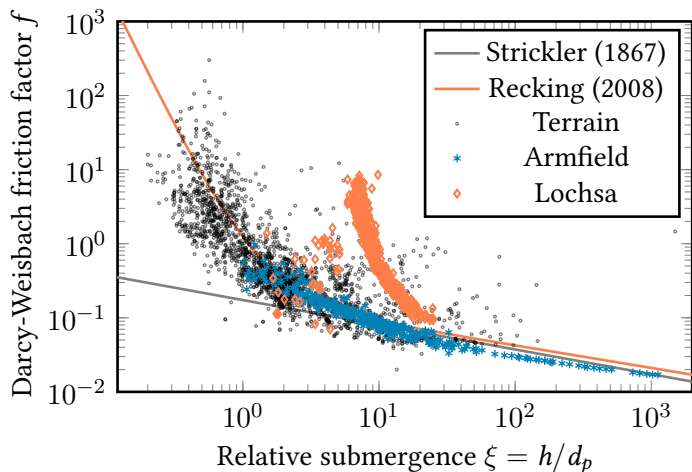
- ρ : the fluid's density,
- g : the gravitational acceleration,
- K : a calibrated coefficient (depends on roughness),
- $\bar{u}$: the mean flow speed and
- h : the flow depth.



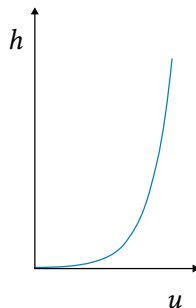
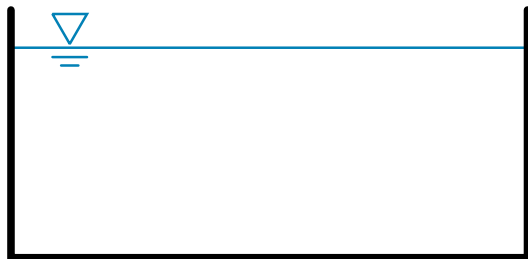
(Gauckler-)Manning-Strickler's validity domain



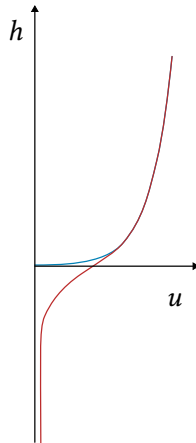
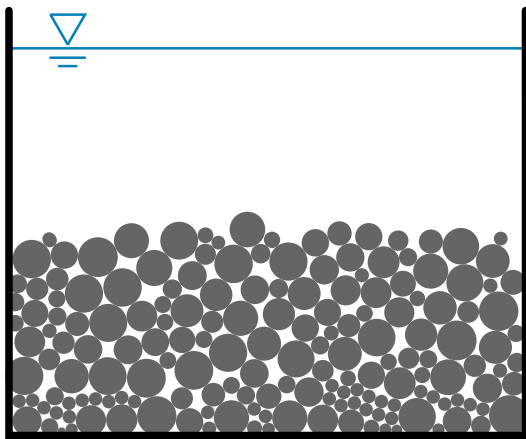
(Gauckler-)Manning-Strickler's validity domain



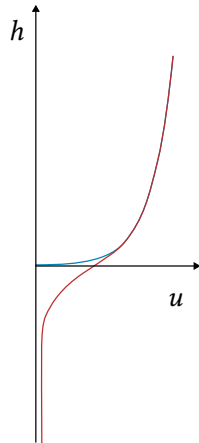
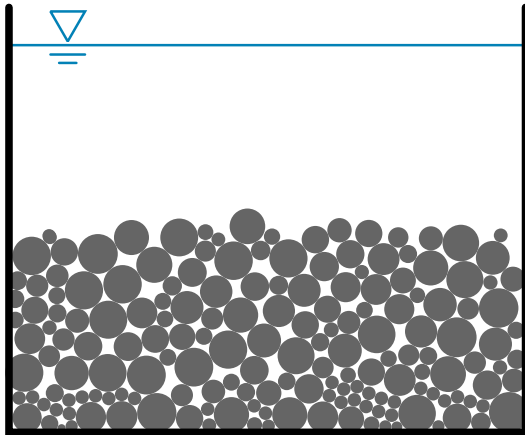
Velocity profile in steep gravel bed rivers



Velocity profile in steep gravel bed rivers

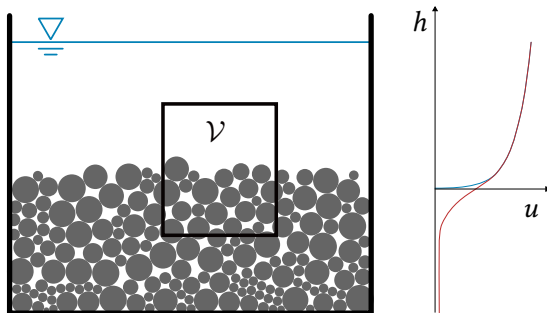


Velocity profile in steep gravel bed rivers



How to include hyporheic flow in the flow resistance?

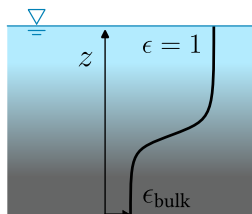
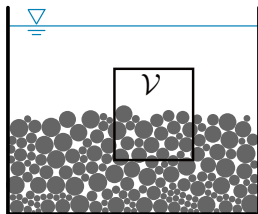
Representative elementary volume



$$u = \int_{\mathcal{V}} \int_T U dt dV$$

$\Rightarrow$ Additional stress in momentum balance : τ_d

Steady-state uniform flow equation



Momentum balance :

$$0 = \epsilon \rho g i + \frac{d(\tau_d + \tau_t + \tau_v)}{dz} + \epsilon f - \rho \mu \frac{d\epsilon}{dz} \frac{du}{dz}$$

Approximate analytical solution

$$0 = \epsilon \rho g i + \frac{d(\tau_d + \tau_t + \tau_v)}{dz} + \epsilon f - \rho \mu \frac{d\epsilon}{dz} \frac{du}{dz}$$

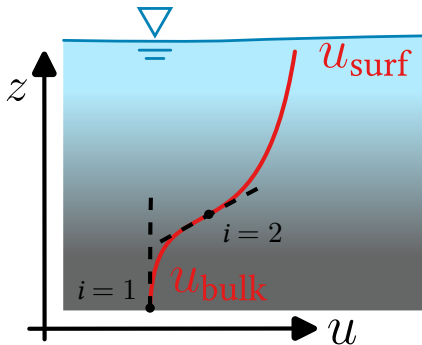
$$1 = c_1 \frac{du}{dz} + c_2 \left(\frac{du}{dz} \right)^3 + c_3 \left(\frac{du}{dz} \right)^4 + c_4 \frac{d^2 u}{dz^2} + c_5 \left(\frac{du}{dz} \right)^2 \frac{d^2 u}{dz^2}$$

$$c_i = c_i \left(u(z), \epsilon(z) \right)$$

Approximate analytical solution

Asymptotic expansions

$$u_i = u_i^{(0)} + \alpha_i u_i^{(1)} + \mathcal{O}(\alpha_i^2)$$



Conclusion

- Hyporheic flow increases the base of the velocity profile.
 - Volume averaging transforms the biphasic media into a continuous porous media.
- ↪ Current step : deriving the analytical approximation of the velocity profile.

References



“Flow Resistance and Hydraulic Geometry in Gravel-And Boulder-Bed Rivers”.



“Du mouvement de l’eau dans les conduites.”.



“Spatially Averaged Open-Channel Flow over Rough Bed”.



“An experimental investigation of turbulent free-surface flows over a steep permeable bed”.